Chicago Housing Risk Analysis: Economic and Housing Structural Determinants of Building Violations

Group Information

Group Number: 46
Course: Section4 – PPHA 30538
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Research Question

How are housing violations distributed across neighborhoods in Chicago, and to what extent can economic pressure and housing structural vulnerability explain these spatial patterns?

Data and Approach

Data Resources

The building violations data come from the City of Chicago Data Portal (`building_violations_raw`), including violation dates, codes/descriptions, and locations.
Census tract–level measures come from the Census Bureau ACS 5-year release (Cook County, IL): `median_household_income.csv` and `rent_burden.csv` (economic pressure), and `poverty_status.csv`, `tenure_owner_renter.csv`, and `units_in_structure.csv` (housing structure).

Data Progress

1. Clean Data: Time window filtering, GEOID standardization, quality checks, build tract-level structure.
2. Further Processing

i. Created Violation Rate: converted raw violation counts into a rate per 1,000 units, equals to 1,000 × (count/total units), ensures that higher values indicate more violations relative to local building stock.

ii. Standardization: Created z-scores Indexes.

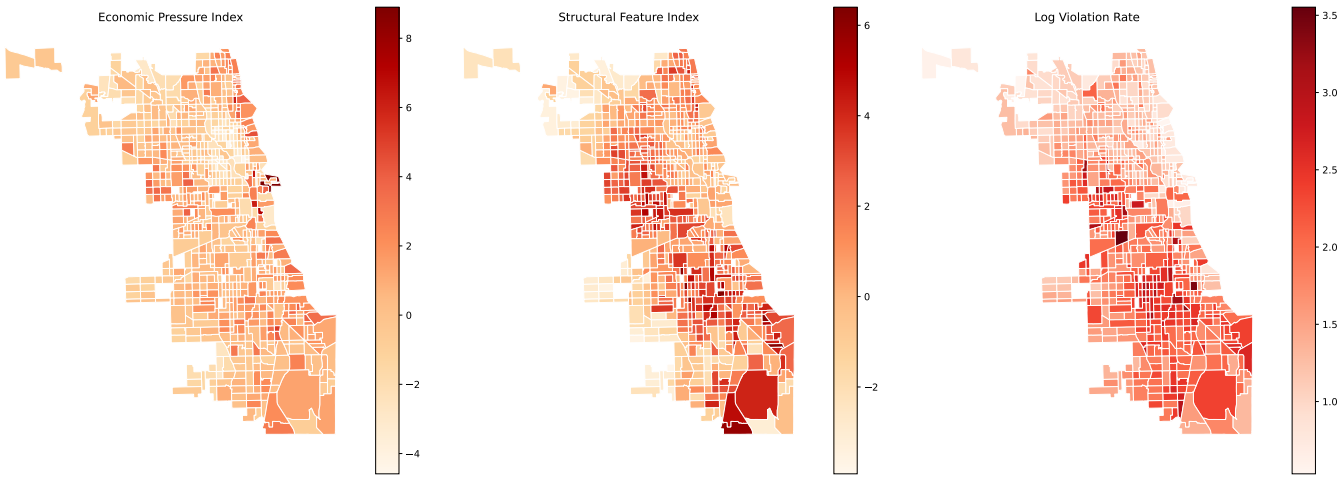
- Economic Pressure Index: rent_burden_z-index (share of households spending 30%) - income_z-index (direction reversa).
- Housing Structural Index: poverty_z-index (share of population below poverty line in the community) + tenure_z-index (share of renters among all households) + housing_age_z-index (share of units built in 1939 or earlier).
  (Here, tenure is measured as the percentage of renter-occupied units in each census tract, rather than tenancy term. Higher renter-occupied share is expected to correlate with higher building violation rates since: Landlords may have weaker incentives for regular maintenance of rental properties compared to owner-occupied homes; Short-term rental tenancy reduces tenants' motivation to report minor violations before they escalate, leading to accumulated housing code violations.)

Weakness and Difficulties

1. Relative rather than absolute measure: Because we use z-score standardization, the index captures relative differences within Cook County rather than absolute levels of housing pressure.
2. Equal weighting assumption: By aggregating standardized variables, we implicitly assume equal contribution of each component, although their real-world impacts on housing pressure may differ.

Static Visualization

Spatial Distribution



Violation rates are clearly spatially clustered, with higher values concentrated in the southern part of the city. Structural vulnerability exhibits a similar pattern of concentration, while economic pressure appears more spatially diffuse.

Correlation Analysis

```
Number of valid data rows after merging: 775
NA check:
GEOID                              0
avg_monthly_total_violation_rate   0
economic_pressure_index            0
housing_structural_index           0
dtype: int64
alt.HConcatChart(...)
```

The plot indicates that housing structural vulnerability is a more consistent predictor of building violation rates than neighborhood economic pressure: right plot shows a moderately stronger positive linear trend between the Housing Structural Feature and average monthly building violation rate.

```
Pearson Correlation:
1. Economic Pressure Index vs Average Monthly Total Violation Rate:
   Correlation coefficient = 0.0840
   p-value = 0.019367
```

```
2. Housing Structural Index vs Average Monthly Total Violation Rate:
   Correlation coefficient = 0.3911
   p-value = 0.000000
```

	economic_pressure_index	housing_structural_index	avg_monthly_total_v
economic_pressure_index	1.000	0.5360	0.0840
housing_structural_index	0.536	1.0000	0.3911
avg_monthly_total_violation_rate	0.084	0.3911	1.0000

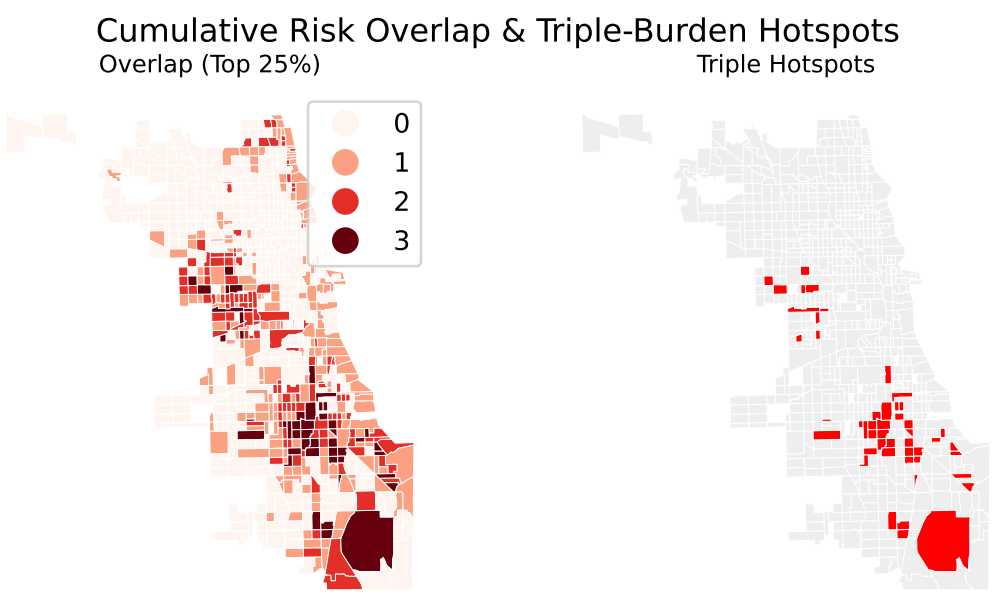
Results of correlation coefficients in the regression matrix indicates that: The weak influence of economic pressure on the rate is likely to be indirectly generated through the mediating variable of housing structural features.

Streamlit

Distribution

We developed an interactive Streamlit dashboard to explore the distribution of housing violations across Chicago census tracts. The dashboard allows users to visualize violation rates, inspect tract-level information, and filter violations by category and time. It also includes a hotspot overlap tool that identifies tracts in the top percentile of economic pressure, structural vulnerability, and violation rates. These exploratory visualizations help identify spatial concentrations of risk and motivate the subsequent analysis of spatial patterns and overlapping housing risks.
Spatially, violation rates show persistent within-city imbalance: hotspots repeatedly cluster in Chicago's southwest corridor, while north-side and lakefront areas remain relatively low.

Policy Extention



The maps show how many risk dimensions each tract falls into within the top 25%. The three pressures overlaps in southern and southwestern Chicago.

These patterns suggest that housing policy should prioritize targeted inspections and maintenance in these areas. Because structural vulnerability is more strongly associated with violations than economic pressure, interventions improving building conditions may be especially effective.